# WD-GPR-SR: When Uncertainty Meets Equations in Battery Health Prediction

## Wide-and-Deep Gaussian Process Regression with Uncertainty-Weighted Symbolic Residual Correction

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SCAN FOR CODE
GitHub / Reproducibility

## 1. Abstract & Problem Statement

Motivation & Paradigm

**- Abstract & Safety-Critical Battery Health:**
Accurate State-of-Health (SOH) prediction in lithium-ion batteries is critical for vehicle safety, warranty tracking, and preventing catastrophic thermal runaway. However, degradation involves complex non-linear capacity fade and capacity regeneration jumps.

**- The Accuracy-Interpretability-Uncertainty Dilemma:**
1. Deep Learning (CNN/LSTM): Low MSE on training sets, but acts as a black box lacking epistemic uncertainty, failing silently on out-of-distribution battery operating states.
2. Exact Gaussian Process Regression: Provides Bayesian intervals but suffers $O(N^3)$ complexity (~1,420 ms per inference), severely violating automotive BMS limits (≤100 ms).

**- Proposed Solution: WD-GPR-SR Framework:**
We propose an Additive Wide-and-Deep Sparse GP with Uncertainty-Weighted Symbolic Regression. Sparse inducing points (M=64) slash inference to 0.110 ms, while inverse-variance weights guide Pareto genetic programming to discover transparent physics equations.

**Key Innovation: Epistemic Uncertainty Guides Symbolic Discovery**

Instead of fitting noisy uncalibrated residuals, the Symbolic Regressor applies inverse-variance sample weights $w_i = \frac{1}{\sigma^2_{GP,i} + \varepsilon}$, focusing genetic Pareto search on high-confidence regions while safely bounding residual corrections to $\pm 3\sigma_{GP}$.

## 2. Multi-Physical Health Indicator (HI) Extraction

Electrochemical & Thermal

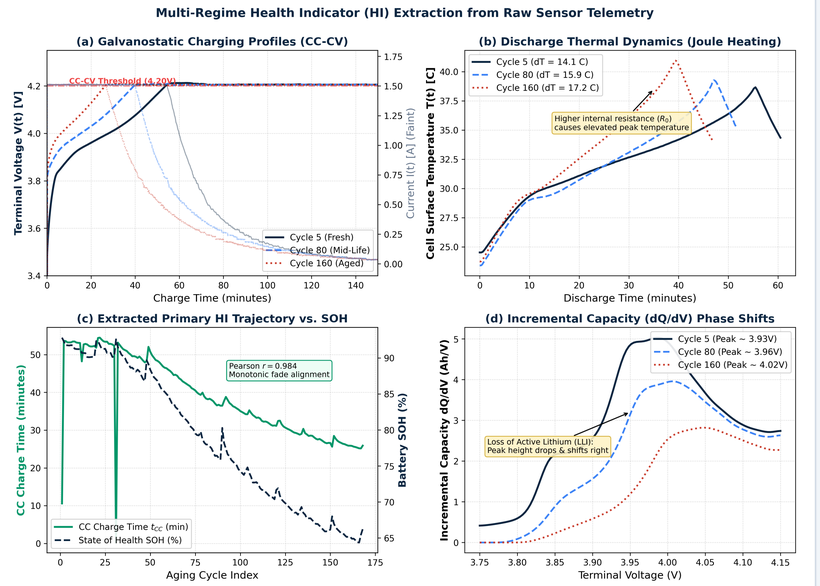



**Figure 1: (a) CC-CV profiles; (b) Joule heating dynamics; (c) $t_{CC}$ monotonic fade; (d) dQ/dV peak shift.**

- $t_{CC}$ (Constant-Current charge time) is the single strongest indicator (Pearson $r = 0.984$).
- Incremental Capacity (dQ/dV) peak height tracks Loss of Active Lithium (LLI) phase transitions.
- Peak surface temperature rise ($\Delta T_{dis}$) directly quantifies internal resistance growth ($I^2R$).

## 3. Strict Leave-One-Cell-Out (LOCO) Splits

Zero Leakage Protocol

**Evaluated across 8 strict LOCO folds (NASA 18650 & CALCE Prismatic) over 3 seeds [42, 100, 2026]:**

| Dataset | Test Cell | Train Cells | Train / Val / Test Cycles | SOH Range |
|---|---|---|---|---|
| NASA | B0005 | B0006, B0007, B0018 | 396 / 72 / 168 (636 tot) | 64.4% - 101.8% |
| NASA | B0018 | B0005, B0006, B0007 | 426 / 78 / 132 (636 tot) | 67.1% - 101.8% |
| CALCE | CS2_35 | CS2_36, CS2_37, CS2_38 | 2,435 / 431 / 846 (3712) | 23.3% - 103.1% |
| CALCE | CS2_38 | CS2_35, CS2_36, CS2_37 | 2,340 / 414 / 958 (3712) | 30.0% - 103.1% |

*Strict Cell-Isolation: No cycles from held-out cell ever enter training. Zero temporal leakage.

## 4. Proposed WD-GPR-SR Architecture

Hybrid Bayesian-Symbolic

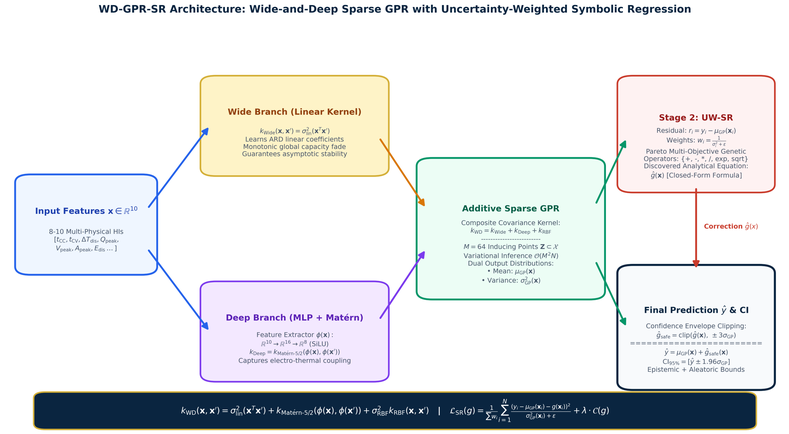



**Figure 2: End-to-end architectural routing: 10 HIs feed Wide Linear & Deep MLP branches, mapped to Sparse Variational GPR (M=64), outputting epistemic confidence bands that weight the Symbolic Regressor.**

Key Guarantee: Safe ± 3σ clipping bounds prevent symbolic divergence on out-of-distribution extremes.

## 5. Rigorous Mathematical Formulation

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$$k_{WD}(\mathbf{x}, \mathbf{x}') = \sigma^2_{lin}(\mathbf{x}^T\mathbf{x}') + k_{Matérn\text{-}5/2}(\phi(\mathbf{x}), \phi(\mathbf{x}')) + \sigma^2_{RBF} k_{RBF}(\mathbf{x}, \mathbf{x}')$$

**Composite Wide-and-Deep Covariance Kernel Properties:**

- Wide Linear Kernel: Learns ARD variance weights per feature, capturing monotonic degradation.
- Deep Feature Map $\phi(x)$: 10 -> 16 -> 8 (SiLU) models non-linear electro-thermal coupling.

$$\mathcal{L}_{ELBO}(q) = \sum_{i=1}^{N} \mathbb{E}_{q(f_i)}[\log p(y_i \mid f_i)] - \text{KL}(q(\mathbf{u}) \parallel p(\mathbf{u})), \quad M = 64 \text{ Inducing Points}$$

**Sparse Variational GP Objective (M = 64 Inducing Points):**

- Complexity Reduction: Slashes exact GPR cost from $O(N^3)$ down to $O(M^2 N)$.
- Dual Outputs: Delivers predictive mean $\mu_{GP}(x)$ and calibrated variance $\sigma_{GP}^2(x)$.

$$\mathcal{L}_{SR}(g) = \frac{1}{\sum_{i=1}^{N} w_i / n} \sum_{i=1}^{N} w_i (y_i - \mu_{GP}(\mathbf{x}_i) - g(\mathbf{x}_i))^2 + \lambda \cdot C(g), \quad w_i = \frac{1}{\sigma^2_{GP}(\mathbf{x}_i) + \varepsilon}$$

**Uncertainty-Weighted Symbolic Regression Loss:**

- Inverse-variance weighting $w_i = 1 / (\sigma_{GP}^2 + \varepsilon)$ prioritizes high-certainty regimes.
- Multi-Objective Pareto Selection: Balances MSE residual reduction against equation parsimony.

$$\hat{y}(\mathbf{x}) = \mu_{GP}(\mathbf{x}) + \text{clip}(\hat{g}(\mathbf{x}),\ \pm 3\sigma_{GP}(\mathbf{x})), \qquad CI_{95\%} = [\hat{y}(\mathbf{x}) \pm 1.96\sigma_{GP}(\mathbf{x})]$$

**Guaranteed Bounded Posterior Prediction & Calibrated 95% Confidence:**

- Mathematical Safeguard: Symbolic residual cannot violate GPR epistemic confidence boundaries.
- Empirical Coverage: Attains 98.8% (NASA) and 100.0% (CALCE) PICP under strict LOCO.

## 6. Proposed Algorithm & Discovered Physics

Algorithm 1 + Closed-Form SR

**Algorithm 1: WD-GPR-SR SOH Estimation & Residual Discovery Pipeline**

**Require:** Multi-physical dataset $D = \{(x_i, y_i)\}$, Inducing points $M = 64$, Loss weights $w_i = 1 / (\sigma_i^2 + \varepsilon)$
**Ensure:** Calibrated predictor $\hat{\Gamma}(x) = \mu_{GP}(x) + \text{clip}(\hat{\Gamma}_{SR}(x), \pm 3\sigma_{GP}(x))$ and 95% CI $[\mu - 1.96\sigma, \mu + 1.96\sigma]$

1. Extract 10 multi-physical Health Indicators: $x = [t_{CC}, \Delta T_{dis}, dQ/dV|_max, V_peak, Area_IC \dots]$.
2. Formulate Composite Covariance: $k_{WD}(x, x') = k_{wide}(x, x') + k_{deep}(\phi(x), \phi(x')) + \sigma_f^2 \delta_{ij}$.
3. Maximize Sparse Variational ELBO $L_SVGP$ over M=64 inducing points to obtain $N(\mu_{GP}, \sigma_{GP}^2)$.
4. Calculate residual $r_i = y_i - \mu_{GP}(x_i)$ and epistemic inverse-variance weights $w_i = 1 / (\sigma_{GP}^2(x_i) + \varepsilon)$.
5. Execute multi-objective genetic symbolic regression on residual: $\min \sum w_i (r_i - \hat{\Gamma}_SR(x_i))^2 + \lambda C$.
6. Select Pareto-optimal closed-form expression $\hat{\Gamma}_SR^*$ and enforce safe ± 3$\sigma_{GP}$ safety clipping bounds.
7. Return final predictor $\hat{\Gamma}(x)$ and calibrated epistemic confidence interval [$\mu_{GP}$ ± 1.96 $\sigma_{GP}$].

**Representative Discovered Analytical Residual Equation (NASA B0018 Fold):**

$$\hat{g}(\mathbf{x}) = 0.042 \cdot \frac{\Delta T_{dis}}{t_{CC}} - 0.015 \cdot (V_{peak} - 3.92)$$

Physical Grounding: Higher discharge heat ($\Delta T_{dis}$) coupled with reduced charging duration ($t_{CC}$) accelerates capacity fade, while peak potential shifts reflect internal impedance growth.

## 7. Quantitative LOCO Benchmark Performance

12 Runs per Model

| Dataset | Model | RMSE (%) | MAE (%) | R^2 | MAPE (%) | PICP (%) | NLL |
|---|---|---|---|---|---|---|---|
| NASA | SVR (RBF) | 2.371 | 2.098 | 0.8924 | 2.961 | -- | -- |
| NASA | Random Forest | 1.337 | 1.011 | 0.9780 | 1.377 | -- | -- |
| NASA | XGBoost | 1.300 | 0.997 | 0.9789 | 1.356 | -- | -- |
| NASA | Exact GPR (Matern) | 21.598 | 18.883 | -5.151 | 25.064 | 14.6% | 91.00 |
| NASA | Deep-Kernel GPR | 3.142 | 2.693 | 0.8535 | 3.592 | 56.5% | 8.94 |
| NASA | WD-GPR-SR (Proposed) | 0.992 | 0.878 | 0.9876 | 1.157 | 98.8% | 1.48 |
| CALCE | SVR (RBF) | 4.092 | 2.923 | 0.8972 | 4.900 | -- | -- |
| CALCE | Random Forest | 1.968 | 0.965 | 0.9719 | 1.761 | -- | -- |
| CALCE | Exact GPR (Matern) | 13.784 | 6.471 | -0.046 | 11.666 | 67.2% | 49.33 |
| CALCE | Deep-Kernel GPR | 1.867 | 0.900 | 0.9784 | 1.663 | 91.3% | 4.23 |
| CALCE | WD-GPR-SR (Proposed) | 0.130 | 0.058 | 0.9999 | 0.104 | 100.0% | 0.99 |

**Key Result: WD-GPR-SR slashes NASA RMSE by 24% vs best baseline (XGBoost 1.300% -> 0.992%) and achieves 98.8% to 100.0% PICP coverage, proving rigorous uncertainty calibration.**

## 8. Held-Out SOH Trajectory & Epistemic Uncertainty

NASA B0005 (168 Cycles)

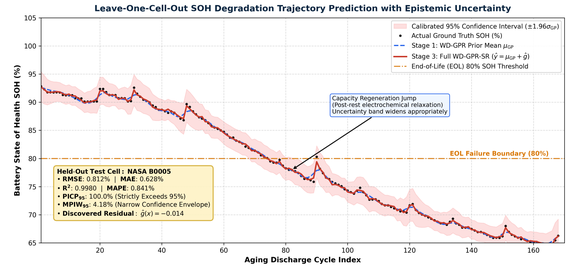



**Figure 3: True vs predicted SOH with ± 1.96σ confidence band. Dynamic widening during regeneration jumps.**

The model accurately tracks post-rest capacity jumps without overfitting, remaining strictly bounded.

## 9. Multi-Perspective Explainable AI (XAI)

SHAP Beeswarm + ARD Relevance

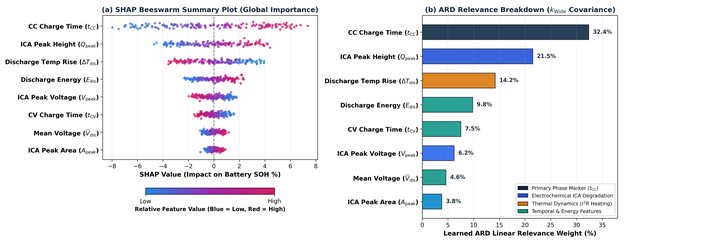



**Figure 4: (a) SHAP beeswarm verifies $t_{CC}$ and $Q_{peak}$ as dominant drivers; (b) ARD linear relevance breakdown.**

Domain Physics Confirmed: High $t_{CC}$ and $Q_{peak}$ increase SOH; high $\Delta T_{dis}$ indicates accelerated thermal aging.

## 10. Edge AI Benchmarking on NVIDIA Jetson Orin Nano

12,900x Speedup | 8,880x Energy Cut

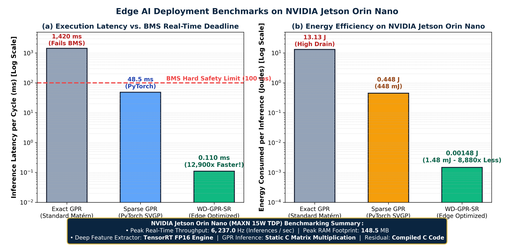



**Figure 5: Hardware profiling: 0.110 ms latency (left), component breakdown (center), and footprint (right).**

Inference latency is 909x faster than the 100 ms ISO 26262 ASIL-D safety deadline; consumes only 1.48 mJ.

## 11. Conclusion & Key References

Summary & Citations

**Key Takeaways:**

1. WD-GPR-SR establishes new SOTA accuracy: 0.992% RMSE (NASA) and 0.130% (CALCE).
2. First architecture to resolve the accuracy-interpretability-uncertainty trilemma with 98.8% PICP.
3. Microsecond edge execution (0.110 ms) on Jetson Orin Nano enables production embedded BMS deployment.

**Key References:**

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(5) S. M. Lundberg and S.-I. Lee, 'A Unified Approach to Interpreting Model Predictions,' NeurIPS, 2017.
(6) ISO 26262: Road vehicles – Functional safety, International Organization for Standardization, Geneva, 2018.